B.Tech. in Electrical Engineering

COURSE PLAN: THEORY COURSE

School Name	**School of Elerctrical Engineering**				
Course Name	**Introduction to Data Science**				
Course Code	**ELE 4449**	**Semester**	**VII**	**Core / PE / OE**	**PE**
Name of the faculty	**Bharathi R. B.**				

No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**0**	**0**	**3**

Mapping of Course Outcomes (COs)/Course Learning Outcomes (CLOs)

CO/ CLO	Statements	No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcomes #	BL* *(Recommended)*
1	Apply Python data analysis librariesand frameworks.	7	22	1,5,7,8,11	1,3,17,18	2,3,4
2	Analyze data using inferential statistics and probability.	8	22	1,2,5,7	1,2,3,17, 18	3,4
3	Analyze data by applying statistical tools and techniques.	8	22	1,2,5,7	1,2,3,17, 18	3,4
4	Analyze data models using supervised learning algorithms of regression & classification.	8	22	1,2,5,7,8,9,11	1,2,3,4,17, 18	3,4,5
5	Analyze data models using dimensionality reduction & unsupervised learning algorithms.	5	12	1,2,5,7	1,2,3,4,17, 18	3,4,5
	Total	**36**	**100**			

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**Preferred BL levels are above 3 (apply)*

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2				3		1	1			1			3
CO2	3	3			3		1							2
CO3	3	3			3		1							2
CO4	3	3			3		1	1	1		1			3
CO5	3	3			3		1							3
Average Articulation Level	**2.8**	**3**			**3**		**1**	**1**	**1**		**1**			**2.6**

Module Matrix#

	LO1	LO2	LO3	LO4	LO5	LO6	LO7	LO8	LO9	LO10	LO11	LO12	LO13	LO14	LO15	LO16	LO17	LO18
CLO1	B		B														B	B
CLO2	B	B	B														B	B
CLO3	B	B	B														B	B
CLO4	B	B	B	B													B	B
CLO5	B	B	B	B													B	B

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Delivery and Assessment Plan of LOs#

Learning Outcome (LO) mapped to the course		Delivery and assessment Plan
LO	**LO statement**	
1	Knowledge and Understanding	Fundamental knowledge of data science concepts, Python programming, data analysis libraries, statistical methods, probability, machine learning, and tutorials, and laboratory Assessment: Assignments, quizzes, laboratory exercises, and written examinations.
2	Engineering Analysis	Analyze datasets using inferential statistics, probability, statistical tools, regression, classification, clustering Assessment: Assignments, quizzes, programming assignments, and written examinations.
3	Design/Development of Solutions	Students develop Python-based data analysis and machine learning solutions by selecting appropriate preprocessing methods, statistical techniques, supervised and unsupervised learning algorithms for different datasets. Assessment: Mini-projects, programming assignments, laboratory assessments, and project evaluation.
4	Investigation	Investigate real-world datasets by collecting, preprocessing, visualizing, analyzing, and interpreting data using statistical and machine learning techniques. Assessment: Laboratory experiments, case studies, project reports.
L17	Planning Skills	Students plan and execute data analysis workflows, organize datasets, select suitable algorithms, manage project timelines, and document the analysis process during laboratory exercises and mini-projects. Assessment: Project planning, laboratory records, assignments, and project evaluation.
L18	Continuous Professional Development	Students develop self-learning skills by exploring emerging Python libraries, machine learning techniques, and recent advancements in data science through independent study, online resources, and project work. Assessment: Self-learning assignments, seminar presentations, project work, and continuous internal assessment.

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Assessment Plan

In – Semester Assessments (MISAC: Mandatory In-Semester Assessment Component and FISAC: Flexible In-Semester Assessment Component)									
S. No.	**Assessment Mode**	**No.**	**Assessment Method**	**Time Duration**	**Marks / Weightage**	**Pattern**	**Typology of Questions**	**Schedule**	**Topics Covered**
1	**MISAC 1**	**1**	**Quiz**	**15 Min**	**5 (5 %)**	**Objective: 5 M** (10 MCQs × 0.5)	Bloom's Level: 2, 3	August 18th, 2026	Covering L1 – L6 **(CO1)**
2	**MISAC 2**	**1**	**Mid-Term Exam**	**120 Min**	**30 (30 %)**	**Objective: 5 M** (10 MCQs × 0.5) **Descriptive: 25 M** (5 x 3 M) + (2 x 4 M) + (1 x 2 M)	Bloom's Level: 2, 3, 4	17th Sep 2026	Comprehensive Examination Covering L1 – L14 Students are expected to answer all questions. **(CO1, CO2)**
3	**MISAC 3**	**1**	**Surprise Assignment**	**20 Min**	**5 (5 %)**	**Descriptive:5M** questions (2M+3 M)	Bloom's Level: 2, 3	Oct 18th ,2026	Covering L1 – L14 **(CO1, CO2)**
4	**FISAC**	**1**	**Mini-Project** (End to End Project with Python – Applying Data Science and Machine Learning)	**4 Weeks**	**10 (10 %)**	Mini-project (Group activity)- Simulation implementation for application-based problems using python programming.	Bloom's Level: 3, 4, 5	September 20th - Oct 24th ,2026	Lecture L1 – L36 **(CO1-CO5)**
End – Semester Assessment									
1	**Regular / Make–up Exam**			**180 Minutes**	**50 (50 %)**	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2 / 3 / 4 / 5 / 6 marks	Bloom's Level: 2, 3, 4,5	27th November 2026	Comprehensive examination covering full syllabus. Students are expected to answer all questions. **(CO 1 – CO5)**

Lesson Plan

L No	Topics	CO Addressed
0	*To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.*	
1	Introduction to the Course	1
2	Steps in Data Analysis, Components & Types of Data Structures, File Formats, Data Sources, ETL	1
3	**Python Libraries & Frameworks:** Python Programming Basics, Python Libraries & Frameworks	1
4	Arrays using NumPy Library	1
5	Linear Algebra and Matrices using NumPy Library	1
6	Working with Data Sets using Pandas Library	1
7	Data Visualization using Matplotlib and Seaborn Python Libraries	2
8	**Inferential Statistics & Probability:** Basic Stats – Mean, Median, Mode, Variance, Standard Deviation	2
9	Standard Deviation, Skewness & Kurtosis, Analytical Problems	2
10	Box and Whisker Plot, Analytical Problems	2
11	Grouped Data Analysis – Analytical Problems	2
12	Probability Basics-PDFs and CDFs	2
13	Discrete Probability Distributions – Bernoulli's, Binomial, Negative Binomial	2
14	Poisson and Geometric Distributions – Python Implementation	2
15	Continuous Probability Distributions – Normal and Standard Normal	3
16	**Statistical Tools & Techniques:** Covariance & Correlation	3
17	Pearsons', Spearman's ad Kendall's Correlation Coefficients	3
18	Bivariate Grouped Data Analysis, Central Limit Theorem and Baye's Theorem	3
19	Confidence Interval and Standard Error: Z and T Distributions	3
20	Hypothesis Testing – Null and Alternate Hypothesis, P-value in Statistics	3
21	Parametric Tests: Z and T Statistical Tests	3
22	Parametric Tests: Goodness of Fit Test, F Statistic and ANOVA Test	3
23	Errors in Hypothesis Testing, Python Implementation of Tests	4
24	**Supervised Data Analytics:** Transformation & Feature Scaling - Attributes, Dimensionality & Filtering	4
25	Linear Regression: Simple & Multiple Linear Regression, Best Fit Line	4
26	Coefficient of Determination, Univariate & Bivariate Distribution Analysis	4
27	Evaluation Metrics, Gradient Descent – Types & Optimizing Cost Function	4
28	Classification: Logistic Regression, Decision Tree Classification	4
29	**Entropy, Gini Impurity, Gini Gain & Gini Coefficient**	4
30	Model Evaluation Metrics – Confusion Matrix, Log Loss Error	4
31	ROC-AUC Curves and PR-AUC Curves Assessment Criteria	5
32	**Dimensionality Reduction and Unsupervised Data Analytics:** Principal Components Analysis	5
33	Time-Series Analysis – Components, Data Types, Stationarity Tests	5
34	**Time-Series Methods – SMA, CMA, EMA, Autocorrelation & Partial Autocorrelation**	5
35	AR, MA, ARMA, ARIMA Time-Series Models	5
36	Clustering Analysis and Association Rule Principles	5
37	Deep Learning Neural Networks Essentials (Self-Directed Learning)	5

Self Directed Learning (SDL)

Sl. No.	TOPIC	No. of learning hours	CO
1	Data types and Functions	2	1
2	*Examples on Python code using Numpy library*	2	1
3	*Examples on Python code using Matplotlib library*	3	1
4	Inferential Statistics	6	2
5	Simple & Multiple Linear Regression	3	2
6	Probability Basics-PDFs and CDFs	3	3
7	Covariance & Correlation	3	3
8	Hypothesis Testing	5	4
9	Logistic Regression	3	4
10	Decision Tree Classification	3	4
11	Random forest classification	3	4
12	Model Evaluation Metrics – Confusion Matrix, Log Loss Error, ROC-AUC Curves	2	4
13	Principal Components Analysis	2	5
14	Time-Series Analysis – Components, Data Types, Stationarity Tests	2	5
15	Time-Series Methods – SMA, CMA, EMA, Autocorrelation & Partial Autocorrelation-examples	2	5
16	spotify-data-analytic	5	5
17	indian-air-quality-analysis-prediction-using-m	5	5
	Total Number of learning hours	**54**	

References:

	Textbooks
1)	Douglas C Montgomery, George C Runger, "Applied Statistics and Probability for Engineers," 5th Edition, John Wiley & Sons, 2011.
2)	Jay L. Devore, "Probability and Statistics for Engineering and the Sciences," Cengage Learning, 2011.
3)	Glenn J. Myatt, Wayne P. Johnson, "Making Sense of Data I: A Practical Guide to Exploratory Data Analysis and DataMining," 2nd Edition, John Wiley & Sons, 2014.
4)	EMC Education Services, "Data Science & Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data," John Wiley & Sons, 2015.
Self-Directed Learning	
Case Studies	https://www.kaggle.com/code/yashwanthn454556/spotify-data-analytics https://www.kaggle.com/code/anbarivan/indian-air-quality-analysis-prediction-using-ml
NPTEL courses	NPTEL Course on "Introduction to Data analytics," by Prof. N. Sudarsanam, Dr. B. Ravindran, IIT Madras. Online (Available): https://nptel.ac.in/courses/110106072.
	SWAYAM Course on "Data Analytics with Python," by Prof. A. Ramesh, IIT Roorkee. Link: https://onlinecourses.nptel.ac.in/noc21_cs45/preview.

Submitted by:

(Signature of the faculty)

Date: 28-07-2026

Approved by:

(Signature of the Dean)